

Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 06

AIM- Time series analysis and forecasting

Theory-Why Time series analysis and forecasting is important?

- To differentiate between random fluctuations or outliers
- To separate genuine insights from seasonal variations.
- To find how data changes over time
- To identify the direction in which the data is changing.

Method

-Select an appropriate dataset for time series forecasting

-Data preparation for time series data

- Change your Date column into date data type and set it as your index.
- Decompose time series data- identify trend ,seasonality and residuals
- Auto correlation Function plot(ACF)
- Partial auto correlation Function plot (PACF)

- Augmented Dickey- Fuller test(ADF)

-Time series forecasting models

- Time series forecasting using linear regression
- Moving average smoothing model (use rolling() function in Pandas)
- AutoRegressive Integrated Moving Average(ARIMA)model
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-Evaluation of models

1. Mean Absolute Error(MAE)
2. Mean Absolute Percentage Error(MAPE)
3. Mean Squared Error(MSE)
4. Root mean square error(RMSE)

Implementaion details

Dataset used

Libraries

- statsmodels.tsa.seasonal
- statsmodels.tsa.stattools.acf
- statsmodels.tsa.stattools.pacf
- statsmodels.tsa.arima.model

